

370445 - (CVE-2007-0981) embedded nulls in location.hostname confuse same-origin checks (Zalewski XSS vulnerability)

370445 - (CVE-2007-0981) embedded nulls in location.hostname confuse same-origin checks (Zalewski XSS vulnerability) - Author not stated, bugzilla.mozilla.org.

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Content

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|

[Third patch + dbaron's comments](#)

19 years ago

[Daniel Veditz [:dveditz]](https://bugzilla.mozilla.org/user_profile?user_id=1689)

4.00 KB, patch

|

bzbarsky

: review+

darin.moz

: review+

dbaron

: superreview+

jay

: approval1.8.1.2+

jay

: approval1.8.0.10+

| [Details](#) | [Diff](#) | [Splinter Review](#) | |

Flags: wanted1.8.1.x+

Flags: wanted1.8.0.x+

Flags: blocking1.8.1.2+

Flags: blocking1.8.0.10+

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[Daniel Veditz [:dveditz]](https://bugzilla.mozilla.org/user_profile?user_id=1689)

Assignee | | |

Updated

19 years ago

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Whiteboard: [sg:high]

Component: Networking: Cookies Networking

QA Contact: networking.cookies networking

Attached patch [Stop hostname attack](#) (obsolete) — [Details](#) — [Splinter Review](#)

[Attachment #255211](#) - Flags: superreview?(darin.moz)

[Attachment #255211](#) - Flags: review?(bzbarsky)

[Attachment #255211](#) - Flags: approval1.8.1.2?

[Attachment #255211](#) - Flags: approval1.8.0.10?

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Summary: Zalewski cookie stealing / same-domain bypass vulnerability Zalewski cookie setting / same-domain bypass vulnerability

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Blocks: [370501](#)

Attached patch [sledgehammer approach to fix ::SetSpec](#) (obsolete) — [Details](#) — [Splinter Review](#)

Assignee: nobody dveditz

Status: NEW ASSIGNED

Attached patch [more limited null squashing in setSpec](#) (obsolete) — [Details](#) — [Splinter Review](#)

[Attachment #255245](#) - Attachment is obsolete: true

[Attachment #255249](#) - Flags: superreview?(jst)

[Attachment #255249](#) - Flags: review?(dbaron)

[Attachment #255249](#) - Flags: approval1.8.1.2?

[Attachment #255249](#) - Flags: approval1.8.0.10?

[Attachment #255245](#) - Attachment is obsolete: false

Attached patch [Third patch + dbaron's comments](#) — [Details](#) — [Splinter Review](#)

? netwerk/base/src/nsStandardURL-1.cpp

Index: netwerk/base/src/nsSimpleURI.cpp

=====

RCS file: /cvsroot/mozilla/netwerk/base/src/nsSimpleURI.cpp,v

retrieving revision 1.43

diff -u -p -6 -r1.43 nsSimpleURI.cpp

--- netwerk/base/src/nsSimpleURI.cpp 29 Jun 2005 21:03:23 -0000 1.43

+++ netwerk/base/src/nsSimpleURI.cpp 15 Feb 2007 21:47:54 -0000

@@ -152,13 +152,13 @@ nsSimpleURI::SetSpec(const nsACString &a

// nsSimpleURI currently restricts the charset to US-ASCII

nsCAutoString spec;

NS_EscapeURL(specPtr, specLen, esc_OnlyNonASCII | esc_AlwaysCopy, spec);

PRInt32 pos = spec.FindChar(':');

- if (pos == -1)

+ if (pos == -1 || !net_IsValidScheme(spec.get(), pos))

return NS_ERROR_MALFORMED_URI;

mScheme.Truncate();

mPath.Truncate();

PRInt32 n = spec.Left(mScheme, pos);

@@ -179,12 +179,18 @@ nsSimpleURI::GetScheme(nsACString &resul

return NS_OK;

}

NS_IMETHODIMP

nsSimpleURI::SetScheme(const nsACString &scheme)

{

+ const nsPromiseFlatCString &flat = PromiseFlatCString(scheme);

+ if (!net_IsValidScheme(flat)) {

+ NS_ERROR("the given url scheme contains invalid characters");

+ return NS_ERROR_MALFORMED_URI;

+ }

+

mScheme = scheme;

ToLowerCase(mScheme);

return NS_OK;

}

NS_IMETHODIMP

Index: netwerk/base/src/nsStandardURL.cpp

=====

```

RCS file: /cvsroot/mozilla/network/base/src/nsStandardURL.cpp,v
retrieving revision 1.82.4.8
diff -u -p -6 -r1.82.4.8 nsStandardURL.cpp
--- netwerk/base/src/nsStandardURL.cpp    22 Jun 2006 19:13:01 -0000 1.82.4.8
+++ netwerk/base/src/nsStandardURL.cpp    15 Feb 2007 21:47:56 -0000
@@ -501,12 +501,14 @@ nsStandardURL::BuildNormalizedSpec(const
    // already point to a [ ] delimited IPv6 address literal.
    // However, perform Unicode normalization on it, as IDN does.
    mHostEncoding = eEncoding_ASCII;
    if (mHost.mLen > 0) {
        const nsCString& tempHost =
            Substring(spec + mHost.mPos, spec + mHost.mPos + mHost.mLen);
+   if (tempHost.FindChar('\0') != kNotFound)
+       return NS_ERROR_MALFORMED_URI; // null embedded in hostname
        if ((useEncHost = NormalizeIDN(tempHost, encHost)))
            approxLen += encHost.Length();
        else
            approxLen += mHost.mLen;
    }

@@ -1405,12 +1407,15 @@ nsStandardURL::SetHost(const nsACString
    if (flat.IsEmpty())
        return NS_OK;
    NS_ERROR("cannot set host on no-auth url");
    return NS_ERROR_UNEXPECTED;
}

+ if (host && strlen(host) < flat.Length())
+     return NS_ERROR_MALFORMED_URI; // found embedded null
+
    InvalidateCache();
    mHostEncoding = eEncoding_ASCII;

    if (!(host && *host)) {
        // remove existing hostname
        if (mHost.mLen > 0) {

```

Index: netwerk/base/src/nsURLHelper.cpp

=====

```

RCS file: /cvsroot/mozilla/network/base/src/nsURLHelper.cpp,v
retrieving revision 1.60.2.3
diff -u -p -6 -r1.60.2.3 nsURLHelper.cpp
--- netwerk/base/src/nsURLHelper.cpp    20 Jul 2006 22:59:11 -0000 1.60.2.3
+++ netwerk/base/src/nsURLHelper.cpp    15 Feb 2007 21:47:57 -0000
@@ -504,17 +504,18 @@ net_ExtractURLScheme(const nsACString &i

```

```

    return NS_ERROR_MALFORMED_URI;
}

PRBool
net_IsValidScheme(const char *scheme, PRUint32 schemeLen)
{
- // first char much be alpha
+ // first char must be alpha
    if (!nsCRT::IsAsciiAlpha(*scheme))
        return PR_FALSE;
-
- for (; schemeLen && *scheme; ++scheme, --schemeLen) {
+
+ // nsCStrings may have embedded nulls -- reject those too
+ for (; schemeLen; ++scheme, --schemeLen) {
    if (!(nsCRT::IsAsciiAlpha(*scheme) ||
        nsCRT::IsAsciiDigit(*scheme) ||
        *scheme == '+' ||
        *scheme == '.' ||
        *scheme == '-')
        return PR_FALSE;

```

[Attachment #255211](#) - Attachment is obsolete: true

[Attachment #255245](#) - Attachment is obsolete: true

[Attachment #255249](#) - Attachment is obsolete: true

[Attachment #255211](#) - Flags: superreview?(darin.moz)

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[Attachment #255211](#) - Flags: approval1.8.1.2?

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[Attachment #255249](#) - Flags: review?(dbaron)

[Attachment #255249](#) - Flags: approval1.8.1.2?

[Attachment #255249](#) - Flags: approval1.8.0.10?

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[Attachment #255252](#) - Flags: superreview?(dbaron)

[Attachment #255252](#) - Flags: review?(bzbarsky)

[Attachment #255252](#) - Flags: approval1.8.1.2?

[Attachment #255252](#) - Flags: approval1.8.0.10?

[Attachment #255252](#) - Flags: review?(bzbarsky) review+

[Attachment #255252](#) - Flags: superreview?(dbaron) superreview+

[Attachment #255252](#) - Flags: approval1.8.1.2?

[Attachment #255252](#) - Flags: approval1.8.1.2+

[Attachment #255252](#) - Flags: approval1.8.0.10?

[Attachment #255252](#) - Flags: approval1.8.0.10+

[Attachment #255252](#) - Flags: review?(darin.moz)

Status: ASSIGNED RESOLVED

Closed: 19 years ago

Keywords: [fixed1.8.0.10](#), [fixed1.8.1.2](#)

Resolution: --- FIXED

[Attachment #255252](#) - Flags: review?(darin.moz) review+

Keywords: [fixed1.8.0.10](#), [fixed1.8.1.2](#) [verified1.8.0.10](#), [verified1.8.1.2](#)

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[Jesse Ruderman](#)

| | |

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Summary: Zalewski cookie setting / same-domain bypass vulnerability embedded nulls in location.hostname confuse same-origin checks (Zalewski XSS vulnerability)

Whiteboard: [sg:high] [sg:high] xss

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Alias: CVE-2007-0981

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[Boris Zbarsky [:bzbarsky]](https://bugzilla.mozilla.org/user_profile?user_id=20209)

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Flags: in-testsuite?

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